

RAG Deployment on AWS

EU AI Act Use Case

What is RAG?

The Problem

LLMs hallucinate. They invent facts, fabricate citations, and their knowledge is frozen at training time.

For high-stakes regulatory text like the EU AI Act, this is not acceptable.

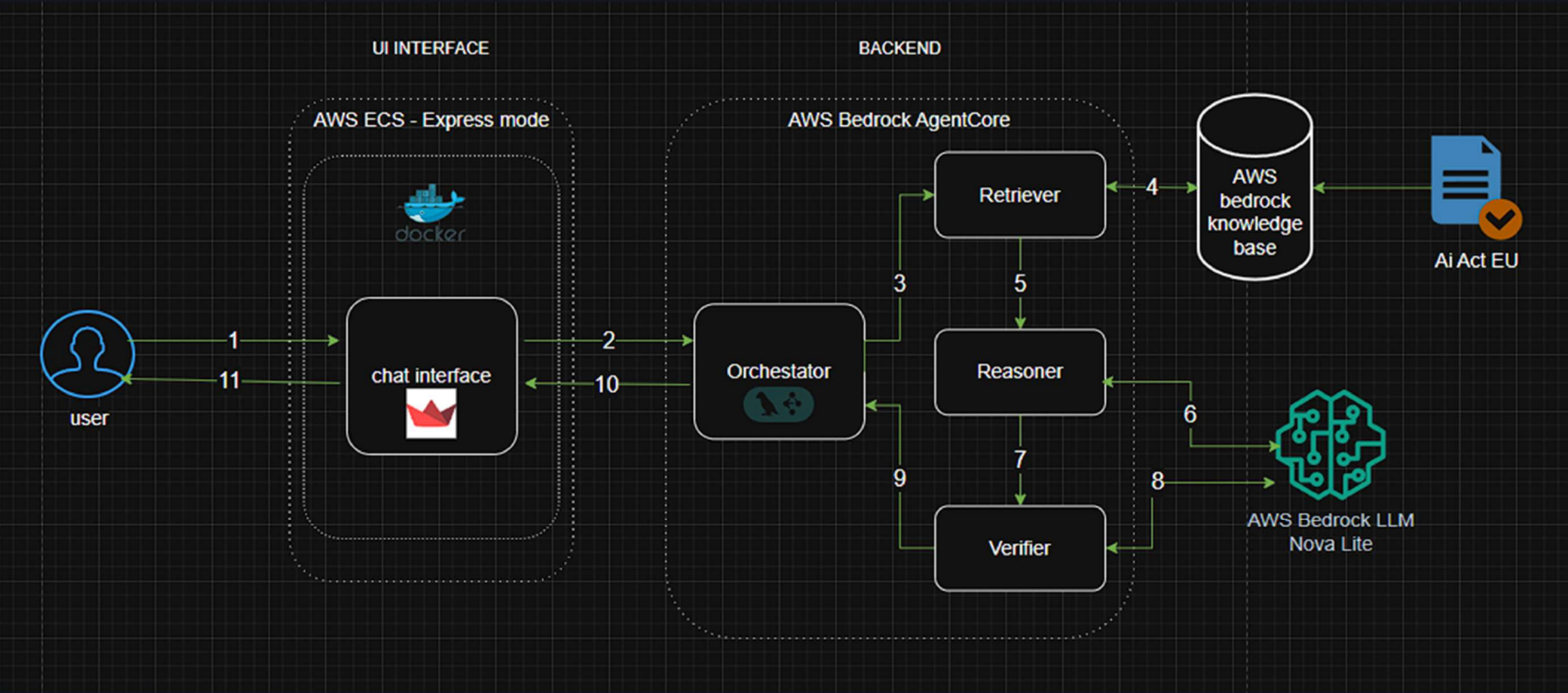
The RAG Solution

Retrieval-Augmented Generation grounds every answer in real text from the document.

Instead of relying on the model's memory, the system retrieves the most relevant passages of the Act, reasons over them, and cites the exact source fragments it used.

Every answer is verifiable. Every claim has a citation.

System Architecture



Technology Stack



AWS Bedrock AgentCore

Serverless agent runtime -- deploys a Docker container on managed infrastructure.



Bedrock Knowledge Bases

Managed vector storage + retrieval -- S3 + Titan Embeddings + OpenSearch.



Amazon Nova Lite

Fast, low-cost multimodal LLM for reasoning via the Bedrock Converse API.



LangGraph

StateGraph orchestrator -- connects agents with typed state and edge routing.



Streamlit

Chat interface on ECS -- calls `invoke_agent_runtime` and shows grounding indicators.



CloudWatch + OTEL

Observability -- logs, traces and metrics across every stage of the pipeline.

Key Takeaways



AgentCore makes deployment trivial

Define your agent in agentcore.json, push a Docker image -- AgentCore handles scaling, routing and IAM.



LangGraph = readable agent logic

StateGraph turns orchestration into a clean DAG. Debugging agents becomes debugging a graph.



Validation is non-negotiable

Without a validator you cannot trust RAG output. Citation checks make hallucinations visible.



Legal documents are the hardest RAG target

Cross-references, precise terminology and citation requirements make the EU AI Act a real test of retrieval quality.

Performance: Measurement & Optimization

How do we measure the performance of a RAG?

How do we tune parameters to improve performance?